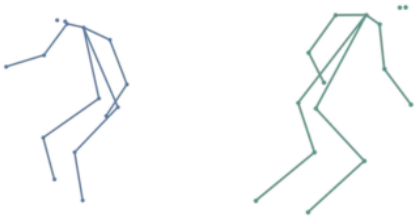
CanonFuse3D: calibration-free fusion of two monocular 3D pose streams

Two monocular 3D pose streams



view L view R

approximately synchronized; no images,
2D keypoints, or camera parameters at fusion

CanonFuse3D

1 Human-centric canonicalization

2 Cross-view pose alignment

3 Cross-view attention +
temporal refinement

4 Joint-frame gate + residual

Refined canonical 3D pose

MPJPE vs. canonical averaging

−44.5% Unity

−48.6% Ski-PTZ-Pose

robust to camera separation,
timing error, and occlusion;
adapts to other pose estimators;
extends to 3–4 cameras